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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EEE211/20EEE211/17EEE211 - PROGRAMMABLE LOGIC CONTROLLERS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat block diagram show the main functional units and ways in which buses link them. Explain briefly the functions of each block.	8	L*2	1	1
b)	Define the following terms in connection with the performance analysis of sensors used in PLC systems.				
	i. Accuracy				
	ii. Sensitivity				
	iii. Stability				
	iv. Repeatability				
	v. Range				
	vi. Reliability	6	L1	1	1
c)	Explain the various signal conditioning circuits used in PLC system.	6	L2	1	1
2. a)	Explain the basic input unit for DC and AC inputs.	8	L2	1	1
b)	Give the steps followed by PLC in processing the inputs connected. What are the factors which will decide scanning time for a PLC.	6	L2	1	1
c)	Mention the basic form of transistor output and input units for both current sinking and sourcing of a PLC unit.	6	L2	1	1
Unit – II					
3. a)	Develop a ladder diagram and write the instructions to enable various sections of the given ladder diagram using Master Control Relays.	8	L3	3	2
b)	Draw the ladder diagrams to illustrate the following functions.				
	i. 2 input AND gate				
	ii. 2 input NAND gate				
	iii. 2 input EXOR gate	6	L1	2	2
c)	Explain the branching and convergence in case of sequential function chart with equivalent ladder diagram.	6	L2	2	3
4. a)	With appropriate connection and ladder diagram, explain the safe position for stop and emergency switches in PLC.	8	L2	2	2
b)	Three fire sensors provide inputs to a SET-RESET function block so that if one of the sensors is activated the alarm is set and remains set until it is cleared by a reset command. Implement this function on a ladder and write the instruction list for the same.	6	L3	3	3
c)	Explain the method of starting multiple outputs using internal relays with ladder diagrams.	6	L1	3	3
5. a)	Explain the One-shot operation with necessary timing and ladder diagram.	8	L2	3	3
b)	Explain various conditional and iterative statements available in structured text programming.	6	L1	2	2

P.T.O.

- c) Draw the functional block diagram and ladder diagram for the case of a valve which is to be operated to lift a load when a pump is running and either the lift switch is operated or an indicative switch deployed to show the already lifted load is operated.

6 L2 2 2

Unit – III

6. a) Explain the different timers as per IEC standards. Using any of those, develop the ladder logic that will turn ON an output light 5 seconds after switch A has been turned ON. Illustrate using timing diagram.
- b) Explain the different data handling functions available in a PLC with example ladder rungs for each.
7. a) Draw the ladder diagrams for the system to carry out following tasks.
- Give an output after a photocell sensor has given 10 pulses input signals as a result of detecting 10 objects passing in front of it
 - Control a machine to direct 6 tins along one path for packaging in a box and then 12 tins along another path for packaging in another box with the help of a deflector.
- b) Explain the operation of a 4 bit shift register using the ladder diagram.
8. a) Draw the ladder diagram for performing the following task:
Bottles are counted until all 12 are in position for filling. When in position all bottles are filled simultaneously for 5s. After filling, there is a delay of 3s for foam to settle. The 12 bottles are then taken out and capped and then counted as they are installed.
- b) Construct the ladder diagram and wiring diagram for a central heating system involving a boiler. Also draw the timing diagram.

10 L3 4 3

10 L2 5 3

10 L4 4 3

10 L2 5 3

10 L4 4 3

10 L2 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
